

Yujie Zhang

Curriculum Vitae

Biography

- 2023 (Dec)- **Postdoctoral fellow**, *University of Waterloo, Institute for Quantum Computing/ Perimeter Institute for Theoretical Physics*, Waterloo, CA.
- 2017 -2023 **Ph. D. in Physics**, *University of Illinois at Urbana-Champaign, Department of Physics*, Urbana, USA.
- 2013-2017 **B. Sc. in Physics**, *Nanjing University, Kuang Yaming Honor School*, Nanjing, China.

Preprints

- 2025 **CritPt group**, *Probing the Critical Point (CritPt) of AI Reasoning: a Frontier Physics Research Benchmark*, arXiv:2509.26574 (2025)
- Y Zhang**, Y Ying, D Schmid, *Quantifiers and witnesses for the nonclassicality of measurements and of states*, arXiv:2504.02944 (2025) [Quantum accepted]

Publications

- 2026 **Y Zhang**, J Spodek, D Schmid, C Reid, T Jennewein, K, Resch, R Spekkens, *Entanglement certification using noncontextuality inequalities*, Phys. Rev. X (2026)
- Y Zhang**, D Schmid, Y Ying, R Spekkens, *Reassessing the boundary between classical and nonclassical for individual quantum processes*, Phys. Rev. X 16, 021050 (2026)
- S Vinet, M Clementi, M Bacchi, **Yujie Zhang** et al, *Time-resolved certification of frequency-bin entanglement over multi-mode channels*, npj Quantum Information 12, 38 (2026)
- 2025 **Y Zhang**, T Jennewein, *Criteria for optimal entanglement-enhanced long baseline imaging protocols*, Phys. Rev. Research 7, 043278 (2025)
- Y Zhang**, J Zhang, E Chitambar, *Cost of Simulating Entanglement in Steering Scenario*, Quantum 9, 1902 (2025)
- S Vinet, W Wu, **Y Zhang**, T Jennewein, *Feasibility study of frequency-encoded photonic qubits over a free-space channel*, Opt. Express 33, 40437-40449 (2025)
- Y Wang, **Y Zhang**, VO Lorenz, *Temporally Localized Quantum Operations on Continuous-Wave Thermal Light*, Phys. Rev. Lett. 135, 113602 (2025)
- Y Wang, **Y Zhang**, VO Lorenz, *Astronomical interferometry using continuous-variable quantum teleportation*, Phys. Rev. Research 7, 023154 (2025)

- K. Kapoor, S. Hoseini, J. Choi, B. E. Nussbaum, **Y Zhang** et.al, *Public Quantum Network: The First Node*, Appl. Phys. Lett. 126 (5): 054002 (2025)
- 2024 **Y Zhang**, E Chitambar, *Exact Steering Bound for Two-Qubit Werner State*, Phys. Rev. Lett. 132, 250201 (2024)
- X Chen*, **Y Zhang***, A Winter, VO Lorenz, E Chitambar, *Information Carried by a Single Particle in Quantum Multiple-Access Channels*, PRA 109, 062420 (2024)
- 2023 K Garay-Palmett, DB Kim, **Y Zhang**, F Domínguez-Serna, VO Lorenz, AB U'Ren, *Fiber-based photon-pair generation: tutorial*, JOSA B **40** (3), 469-490 (2023)
- D Diaz, **Y Zhang**, Y Wang, VO Lorenz, P Kwiat, *Toward Quantum-enhanced Interferometric telescoping*, Frontiers in Optics (2023)
- 2022 **Y Zhang**, X Chen, E Chitambar, *Building multiple access channels with a single particle*, Quantum **6**, 653 (2022)
- 2021 Y Wang, **Y Zhang**, VO Lorenz, *Superresolution in interferometric imaging of strong thermal sources*, Phys. Rev. A **104**, 022613 (2021)
- 2020 **Y Zhang***, RA Bravo*, VO Lorenz, E Chitambar, *Channel activation of CHSH nonlocality*, New Journal of Physics 22 043003 (2020)
- 2019 **Y Zhang**, R Spiniolas, K Shinbrough, B Fang, O Cohen, and VO. Lorenz, *Dual-pump approach to photon-pair generation: demonstration of enhanced characterization and engineering capabilities*, Opt. Express **27**, 19050-19061 (2019)
- 2018 A Zhang, **Y Zhang**, F Xu, L Li, and L Zhang, *Adaptive tomography of qubits: purity versus statistical fluctuations*, J.Phys.A: Math.Theor. **51** 395304 (2018)

Talks

- 2026 **YJ Zhang**, et al, "A paradigm for Entanglement certification using noncontextuality inequalities " Foxconn
- 2025 **YJ Zhang**, et al, "Criteria for optimal entanglement-enhanced long baseline imaging protocols" AQIS
- YJ Zhang**, et al, "Defining nonclassicality for individual quantum process" QPL
- 2024 **YJ Zhang**, et al, "Exact steering bound for two-Qubit Werner state," QIP
- 2023 **YJ Zhang**, et al, "Quantum enhanced interferometric imaging" Quantum 2.0
- 2022 XA Chen*, **YJ Zhang***, et al, "Information carried by a single particle in multiple-access channels" TQC
- 2021 **YJ Zhang**, et al, "Building multiple-access channel with one particle" Beyond IID
- 2020 **YJ Zhang**, et al, "Dual-Pump design enables novel photon-pair characterization and engineering" CLEO
- 2019 **YJ Zhang**, et al, "Tailored photon-pair generation in optical fiber through dual-pump spontaneous four-wave mixing" FIO-LS

* Co-first authors

Awards and Honors

2023 **Boeing Quantum Creators Prize.**

The Boeing Quantum Creators Prize ([Link](#)) recognizes early-career researchers for work that moves the field of quantum information science and engineering in new directions. The program began in 2021 and expanded in 2023 thanks to a commitment from Boeing. Prizes are awarded at the annual Boeing Quantum Creators Prize Symposium hosted during the annual Chicago Quantum Summit.

2019 **University Fellowship for outstanding research accomplishments.**

This graduate fellowship is awarded to a student who has demonstrated significant ability in research.

Outreach

2023 **Public Quantum Network at Urbana Free Library.**

Leading the creation of the Public Quantum Network ([Link](#)) that distributes entangled photons from UIUC to a local library, integrating research and education.

2023 **IQUIST at Ant-Man The Wasp: Quantumania.**

Led a post-screening public discussion on quantum science after Ant-Man and the Wasp: Quantumania; event coordinated by the Illinois Quantum Information Science and Technology Center (IQUIST).